

Technical Note: Exact Octic Operator Hierarchy from Linear to Non-Linear Rotors

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Scope

This note records the exact operator-level results obtained for the octic rotational sector. The goal is to isolate the part of the theory that is already rigorous independently of the unresolved compact formulas for the vibrational dependence of quartic centrifugal distortion constants.

1. Linear audit of Paper 1 versus Paper 2

The symbolic audit of the printed Watson–Aliev source equations and of the compact Aliev linear formulas gives the following exact conclusions:

1. the compact channel V is derivable from the printed bare H_{24} block;
2. the compact channel U is not derivable from the printed form of Paper 1;
3. both $\beta_{\parallel}$ and $\beta_{\perp}$ depend essentially on U ;
4. therefore the compact formulas of Paper 2 for $\beta_{\parallel}$ and $\beta_{\perp}$ are not derivable symbolically from the printed equations of Paper 1;
5. the non-compact equations currently retained in the code coincide with what is derivable from the printed form of Paper 1.

This isolates the unresolved problem to the compact U channel and removes it from the octic branch.

2. Exact octic operator hierarchy

Let $\mathcal{O}_8$ be the commuting rotational operator space of total degree eight in J_x, J_y, J_z .

2.1 Asymmetric top

For an asymmetric top the commuting octic basis has dimension

$$\dim \mathcal{O}_8^{\text{asym}} = 15.$$

The general operator is

$$\mathcal{O}_8^{\text{asym}} = \sum_{a+b+c=4} c_{2a,2b,2c} J_x^{2a} J_y^{2b} J_z^{2c}.$$

2.2 Symmetric top

For a symmetric top with axis a the operator space collapses to dimension

$$\dim \mathcal{O}_8^{\text{sym}} = 5$$

with basis

$$J^8, \quad J^6 J_a^2, \quad J^4 J_a^4, \quad J^2 J_a^6, \quad J_a^8.$$

Writing the corresponding coefficients as

$$L_J^{(4)}, \quad L_{J^3 K}, \quad L_{J^2 K^2}, \quad L_{JK^3}, \quad L_K^{(4)},$$

the exact projection from the cartesian commuting coefficients is

$$L_J^{(4)} = c_{080},$$

$$L_{J^3 K} = -4c_{080} + c_{260},$$

$$L_{J^2 K^2} = 6c_{080} - 3c_{260} + c_{440},$$

$$L_{JK^3} = -4c_{080} + 3c_{260} - 2c_{440} + c_{620},$$

$$L_K^{(4)} = c_{080} - c_{260} + c_{440} - c_{620} + c_{800}.$$

The exact cartesian image constraints for the a -axis symmetric-top subspace are

$$c_{422} = 2c_{440}, \quad c_{242} = 3c_{260}, \quad c_{062} = 4c_{080}, \quad c_{044} = 6c_{080},$$

together with the obvious $b \leftrightarrow c$ equalities

$$c_{602} = c_{620}, \quad c_{404} = c_{440}, \quad c_{224} = c_{242}, \quad c_{026} = c_{062}, \quad c_{008} = c_{080}.$$

2.3 Linear rotor

In the linear limit only one commuting degree-eight operator survives:

$$\dim \mathcal{O}_8^{\text{lin}} = 1.$$

For axis a this operator is

$$(J_y^2 + J_z^2)^4 = J_y^8 + 4J_y^6 J_z^2 + 6J_y^4 J_z^4 + 4J_y^2 J_z^6 + J_z^8.$$

Therefore the exact linear constraints on the cartesian coefficients are

$$c_{800} = c_{620} = c_{602} = c_{440} = c_{422} = c_{404} = c_{260} = c_{242} = c_{224} = c_{206} = 0,$$

$$c_{080} = L, \quad c_{062} = 4L, \quad c_{044} = 6L, \quad c_{026} = 4L, \quad c_{008} = L.$$

Equivalently,

$$L = c_{080} = L_J^{(4)}.$$

3. Consequence for the next derivation step

The operator algebra is therefore closed. The remaining theoretical problem is not the projection machinery but the derivation of the cartesian octic coefficients $c_{2a,2b,2c}$ from the general Watson–Aliev Hamiltonian.

At the current stage the rigorous target is:

$$\text{Paper 1} \longrightarrow c_{2a,2b,2c} \longrightarrow (L_J^{(4)}, L_{J^3 K}, L_{J^2 K^2}, L_{JK^3}, L_K^{(4)}) \longrightarrow L.$$

4. Data requirements

For this octic branch there is currently no evidence that third derivatives of the inverse inertia tensor are needed. The natural tensor hierarchy is instead the same one already identified for the quartic rotational tensor:

$$\mu_1, \quad \mu_2, \quad \Phi_3.$$

In particular, μ_3 is not required by the operator analysis developed so far.